

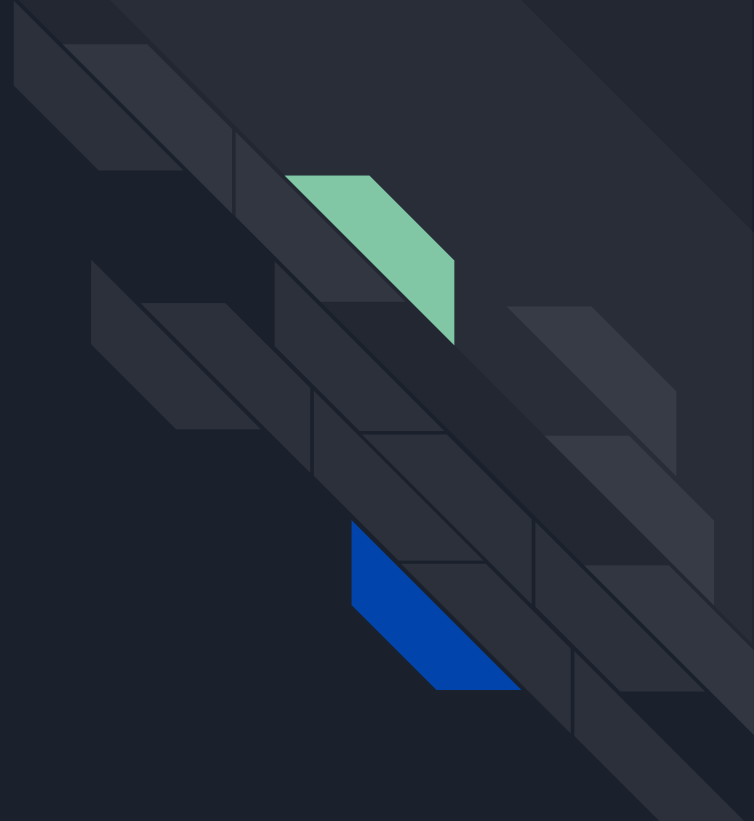
A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with faint, lighter blue diagonal stripes.

Prototype Presentation

Team details

Team name: Snow

Name: Nikitha S





Theme overview

Theme: Art, Culture and Tourism: Bridging art, culture and tourism with tech

Project name: Heritage Visualizer and Analytics Dashboard

Description: The project addresses the challenge of under-represented cultural heritage data by creating an interactive map and analytics dashboard for India's heritage sites.

Objective: To provide a powerful yet simple tool to explore, search, and analyze historical sites across the state using real-time geospatial and categorical insights.



Idea and approach

- 01 Collect data available from data.gov.in
- 02 Combine and process these data
- 03 Upload to snowflake tables and query them when user interacts with Streamlit app
- 04 Using google api fetch details about crowd rate and promote places that are less visited by tourists (yet to be implemented)

Technical Stack

Frontend: Streamlit

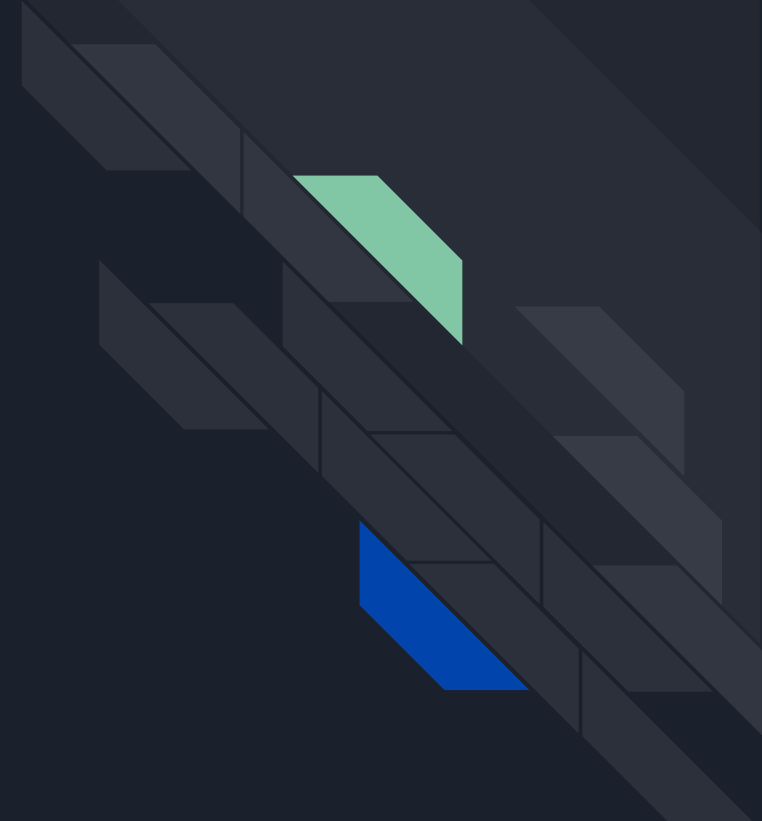
Data Platform: Snowflake

Data Source: <https://data.gov.in> -
for heritage sites of each state

Visualization: matplotlib

Data Format: Excel/CSV

Version Control: GitHub (for
project versioning and open-source
collaboration)



Innovative highlights

1. **Personalized Story Experience**

The application allows users to generate creative stories based on their travel destinations or preferences

(Future Scope)

2. **AI-Powered Place Discovery**

In the next phase of development, the app will integrate **Google Places API** and **Google Maps Traffic/Reviews data** to analyze **lesser-visited or underrated travel spots**. These will be intelligently suggested to users based on:

- Low current visit rates (from Maps popularity data)
- High potential interest (based on reviews, photos, or tags like "hidden gem", "local favorite")
- Geographic proximity to user's input or selected destinations





Feasibility and User-Friendliness:

The application is built using Streamlit and Snowflake, ensuring a smooth and visually engaging experience without the need for any technical expertise. The user interface is intuitive and accessible to a wide audience, including casual travelers, local tourism boards, and cultural researchers.

Key features like interactive storytelling, data-driven visualization, and future integration with Google APIs make it both robust and user-friendly. The backend leverages Snowflake for scalable, efficient data handling, while Streamlit enables rapid development and easy deployment.

SUCCESS METRICS

- **Engagement Rate:**
Number of users generating and exploring AI-generated travel stories and visualizations.
- **Discovery Index (*Post-Google API Integration*):**
Number of lesser-known or under-visited places suggested and selected by users.
- **Data Scalability:**
Ability to plug in new datasets — such as seasonal travel data, regional art forms, or real-time location analytics — with minimal rework.
- **Geo-Diversity Score:**
Range of unique locations and cultural elements represented across user-generated content.





Methodology



Data Collection:

Cultural data was sourced from data.gov.in and converted from CSV to Excel format for initial readability and processing.

Data Transformation & Storage:

Using Python, the data was cleaned, normalized, and uploaded to Snowflake using the `snowflake-connector-python`. This ensures scalable, structured storage with easy SQL-based querying.

Data Exploration & Filtering:

The Streamlit frontend connects to Snowflake and provides filtering options such as region-wise, art-form-wise, and season-wise views.

Story Creation & Visualization:

Streamlit renders interactive maps using `pydeck`, visual plots with `matplotlib`, and cultural timelines to generate engaging narratives.

Methodology

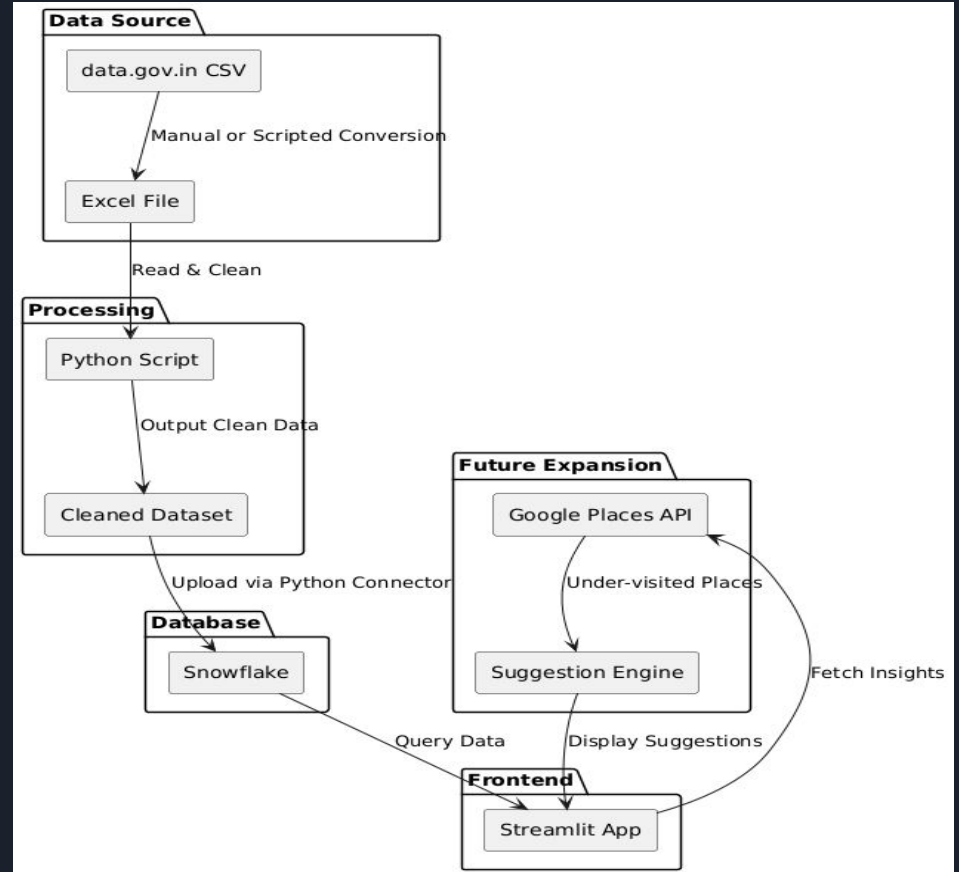
(cont..)

Deployment:

The app is deployed using **Streamlit Cloud** for quick access and sharing.

Feedback Loop (Future Enhancement):

Planned integration of **Google Places API** to analyze under-visited locations and surface hidden cultural gems. Community and local artist input mechanisms are also planned.





Open source Disclosure

- **Streamlit**

Frontend web app development

- **Snowflake (Free Tier)**

Cloud data storage and querying

Snowflake Trial Agreement / Free Developer License

- **matplotlib**

Data visualization libraries

- **Data.gov.in**

Open government datasets

Government Open Data License - India



Observations:

These are the things I observed while creating this project:

- The data related to the Heritage sites of India are mostly outdated.
- There are many states without any data, this includes:
Goa, Haryana, Himachal Pradesh, Jharkhand, Meghalaya, Mizoram, Punjab, Telangana, Tripura, Uttarakhand, West Bengal
- Even the states which had data did not include all the significant places of heritage and importance

Note: I worked on this project as a solo developer so the chances that I might have missed data exists, in such case I would appreciate if anybody who has data regarding the places location can contact me so that I can expand the project.
Thank you.



Prototype Demonstration

Demo Link:

Web app running: <https://youtu.be/tFqKoFghHFA>

Code and DB : <https://youtu.be/GbNzzL6uhT8>

Deployment Link:

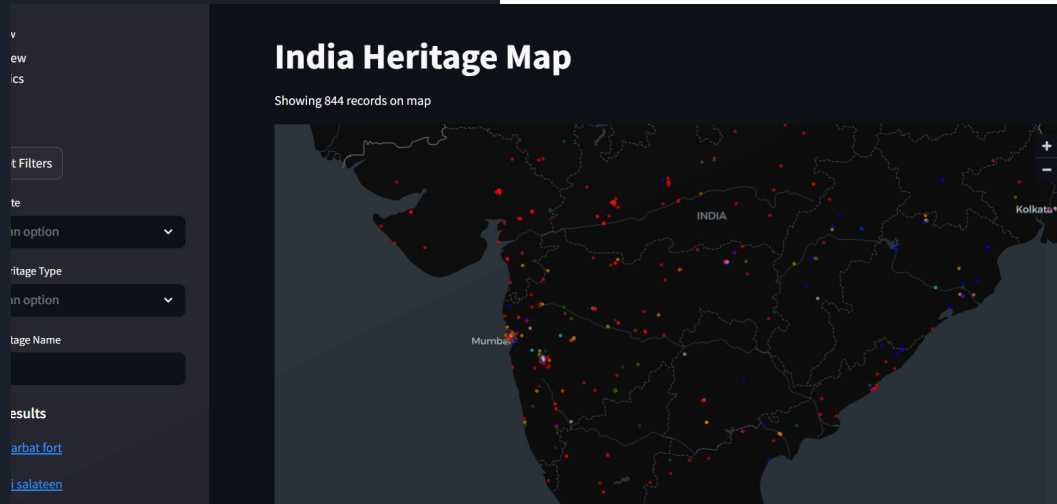
<https://your-story-challenge-8re2xtuchycke5f2dtgzs2.streamlit.app/>

Source Code Repository: <https://github.com/stargalax/Your-story-challenge>

README Instructions:

<https://github.com/stargalax/Your-story-challenge/blob/Main/README.md>

Spotlight on web app



THANK YOU